

E-Commerce Transactions

Customer Segmentation (Clustering) Report

Customer segmentation was performed to group customers with similar purchasing behavior. This helps businesses with:

- Personalized marketing.
- Customer retention strategies.
- Improved product recommendations.

Methodology

1. Feature Selection: Key behavioral attributes such as total spend, average purchase value, and transaction frequency were used for clustering.
2. Scaling: Data was standardized to ensure equal weighting across different attributes.
3. Clustering Algorithm: K-Means clustering was applied to segment customers into groups.
4. Evaluation: The Davies-Bouldin Index was used to assess the quality of clustering.

DB Index value we got from our model is

Davies-Bouldin Index: 0.8524813520458038

Results

Customers were segmented into five distinct groups. A scatter plot visualization of clusters revealed distinct spending patterns, which can guide tailored business strategies.

